

Information Retrieval - Project

Chenliang Li

School of Cyber Science and Engineering,

Wuhan University

Text Search

- Write from scratch a custom command line search engine to index 37,600+ TDT3 news articles
- Implementation
  - Any programming language of your choice: e.g., Assembly, C#, C, Java, Python …..
  - Can use **Lucene (Recommended)** to build the index (A tutorial about Lucene will be provided)
  - implement the document relevance score function **by yourself**
- Preprocessing
  - Perform or ignore stemming
  - Distinguish or ignore upper/lower case characters
  - Process or ignore numbers
  - Process or ignore punctuations including hyperns

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Text Search

- Top-N (configurable) results should be returned via the console/screen along with rank, scores, docID, and summary snippets in the following suggested format

```
q1 = hurricane (took 0.1 seconds)

01 [0.3330] NYT19981231_0439
Liberated at last from a life of revolution, two Khmer Rouge leaders came to
this beach resort Thursday with their families, their leisure wear and their
sunglasses. But their holiday was a drab one, spent mostly behind the dirty
brown curtains of their tiny hotel rooms, ...

02 [0.1234] NBC19981231.1830.0424
With a Gallup survey out tonight showing the president the most admired man in
America, ranking ahead of the Pope, the recently impeached Mr. Clinton is
vacationing in Hilton head ...

...

10 [0.003] PRI19981231.2000.1867
Just days before the Senate is set to start President Clinton's trial, a
proposal has emerged that may avoid a full scale hearing. The trial over the
Monica Lewinsky affair co...
```

- N is configured via a command line option --hits XX, e.g. to return top 27 results, type

```
#search --hits=27 new york san francisco
```

Text Search

➢ Plain Queries

- Your system should support simple command line queries
  - Free Text (order independent):
    # search new york san francisco
  - Doubly quoted phrase queries mixed with free text queries:
    # search “new york city” apple cider
- The following 5 queries (testing queries) into your search engine:
  - Q1 – hurricane
  - Q2 – mitch george
  - Q3 – bill clinton israel
  - Q4 – “newt gingrich” down
  - Q5 – nba strike closed-door
- Include your top 10 screen-shot with the results in your report

Text Search

- Only one student per group
  - One technical report (words/PDF)
    - Including source.zip
- Deadline (roughly one month)
  - Deadline = 2026.12.12 - 23:59
  - **Early submissions are encouraged with extra credits**
  - **Email: 1468572917@qq.com**

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其他

- Lucene部分相关资料
  - https://baike.baidu.com/item/Lucene/6753302?fr=aladdin
  - https://lucene.apache.org/
  - https://lucene.apache.org/core/
  - https://lucene.apache.org/core/8_11_2/demo/overview-summary.html#overview.description
  - https://blog.csdn.net/qq_23853743/article/details/107189544
  - https://zhuanlan.zhihu.com/p/387916586
  - https://www.bilibili.com/video/av262734868/

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